

Auteur: Stan Van Campenhout
 Studierichting: Informatica
 Oefeningenreeks 3H nr. 9.1

$$r = 1 + 2 \sin \theta$$

$$\text{dom} = \mathbb{R}$$

$$p = 2\pi$$

$$\rightarrow \text{BD} = [0, 2\pi]$$

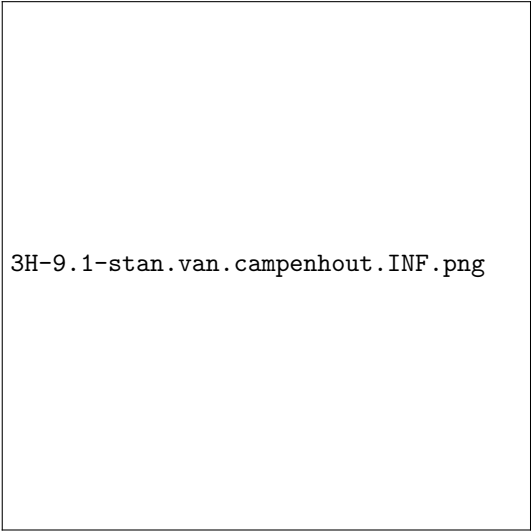
$$r = 0 \Leftrightarrow 2 \sin \theta = -1 \Leftrightarrow \sin \theta = -\frac{1}{2} \Leftrightarrow \theta = \frac{7\pi}{6} + k\pi, \frac{11\pi}{6} + k\pi, \forall k \in \mathbb{Z}$$

$$\rightarrow \left\{ \frac{7\pi}{6}, \frac{11\pi}{6} \right\}$$

$$r' = 0 \Leftrightarrow 2 \cos \theta = 0 \Leftrightarrow \theta = \frac{\pi}{2} + k\pi, \forall k \in \mathbb{Z}$$

$$\rightarrow \left\{ \frac{\pi}{2}, \frac{3\pi}{2} \right\}$$

θ	0		$\frac{\pi}{2}$		$\frac{7\pi}{6}$		$\frac{3\pi}{2}$		$\frac{11\pi}{6}$		2π	
r	+	+	+	+	0	-	-	-	0	+	+	+
r'	+	+	0	-	-	-	0	+	+	+	+	+
r	$\nearrow$	$\nearrow$	M	$\searrow$	$\searrow$	$\searrow$	m	$\nearrow$	$\nearrow$	$\nearrow$	$\nearrow$	$\nearrow$
			3				-1					



References